

For the Mecanum-wheeled robot Connection Pinout Table

This document outlines the pin connections for four motors connected via two motor drivers and an Arduino Mega with WiFi.

Motor	M+ (Motor +)	M- (Motor -)	VCC (Sensor +5V)	A (Sensor A phase)	GND (Sensor GND)
Motor 1	Motor Driver 1 - OUT1	Motor Driver 1 - OUT2	Arduino Mega Pin 31	Arduino Mega Pin 21	Arduino Mega Pin 30
Motor 2	Motor Driver 1 - OUT3	Motor Driver 1 - OUT4	Arduino Mega Pin 33	Arduino Mega Pin 20	Arduino Mega Pin 32
Motor 3	Motor Driver 2 - OUT1	Motor Driver 2 - OUT2	Arduino Mega Pin 35	Arduino Mega Pin 19	Arduino Mega Pin 34
Motor 4	Motor Driver 2 - OUT3	Motor Driver 2 - OUT4	Arduino Mega Pin 37	Arduino Mega Pin 18	Arduino Mega Pin 36

Motor Driver Power Connections:

- Motor Driver 1 and Motor Driver 2 are connected together via 9V and GND.
- 9V line connected to Arduino Mega VIN.
- GND line connected to Arduino Mega GND.
- ENA and ENB pins of the motor drivers are shorted.

Motor Driver IN1-IN4 Pin Connections

Motor Driver	IN Pin	Arduino Mega Pin
Motor Driver 1	IN1	Arduino Mega Pin 4
Motor Driver 1	IN2	Arduino Mega Pin 5
Motor Driver 1	IN3	Arduino Mega Pin 6
Motor Driver 1	IN4	Arduino Mega Pin 7
Motor Driver 2	IN1	Arduino Mega Pin 9
Motor Driver 2	IN2	Arduino Mega Pin 10
Motor Driver 2	IN3	Arduino Mega Pin 11
Motor Driver 2	IN4	Arduino Mega Pin 12